Q3(b)(ii): Corner Solution Explained When RHS < LHS at $x = 100$

ECO 310

The Setup

You have \$100 to invest between two assets:

- Asset 1: invest x dollars
- Asset 2: invest $(100 - x)$ dollars

Three states of the world:

- State A: probability 1/3, returns (r_1^A, r_2^A)
- State B: probability 1/3, returns (r^B, r^B) (both assets same)
- State C: probability 1/3, returns (r_1^C, r_2^C)

Expected utility:

$$EU(x) = \frac{1}{3}u(w_0 + r_1^A x + r_2^A(100 - x)) + \frac{1}{3}u(w_0 + 100r^B) + \frac{1}{3}u(w_0 + r_1^C x + r_2^C(100 - x))$$

First-Order Condition

Taking the derivative with respect to x:

$$EU'(x) = \frac{1}{3}(r_1^A - r_2^A)u'(w^A) + \frac{1}{3}(r_1^C - r_2^C)u'(w^C)$$

where:

- $w^A = w_0 + r_1^A x + r_2^A(100 - x)$ (wealth in state A)
- $w^C = w_0 + r_1^C x + r_2^C(100 - x)$ (wealth in state C)

Simplifying:

$$EU'(x) = \frac{1}{3}\left[(r_1^A - r_2^A)u'(w^A) + (r_1^C - r_2^C)u'(w^C)\right]$$

Multiply by 3 and rearrange:

$$EU'(x) = (r_1^A - r_2^A)u'(w^A) - (r_2^C - r_1^C)u'(w^C)$$

Interior Optimum: Equality Holds

At an **interior solution** where $0 < x^* < 100$, we set $EU'(x^*) = 0$:

$$(r_1^A - r_2^A)u'(w^A) = (r_2^C - r_1^C)u'(w^C)$$

Rearranging:

$$\frac{u'(w^C)}{u'(w^A)} = \frac{r_1^A - r_2^A}{r_2^C - r_1^C} \qquad (*)$$

At an interior optimum: LHS = RHS
This balances the marginal benefit of investing in asset 1 vs asset 2.

Corner Solution: Inequality Holds

But what if equation (*) **cannot be satisfied** for any $x \in [0, 100]$?

Case 1: RHS $<$ LHS even at $x = 100$

$$\frac{r_1^A - r_2^A}{r_2^C - r_1^C} < \frac{u'(w^C)}{u'(w^A)} \quad \text{even when } x = 100$$

What does this mean?

This inequality tells us that even at $x = 100$:

$$EU'(x) = (r_1^A - r_2^A)u'(w^A) - (r_2^C - r_1^C)u'(w^C) > 0$$

Even at $x = 100$, the derivative is positive!

What Should the Decision-Maker Do?

If $EU'(x) > 0$ even at $x = 100$:
The decision-maker wants to **increase** x even more.
Optimal solution: $x^* = 100$
Invest **everything** (\$100) in asset 1, invest **nothing** in asset 2.

Why?

Interpretation of $EU'(x) > 0$:

The derivative tells us: "If I increase x by one dollar, my expected utility goes **up**."

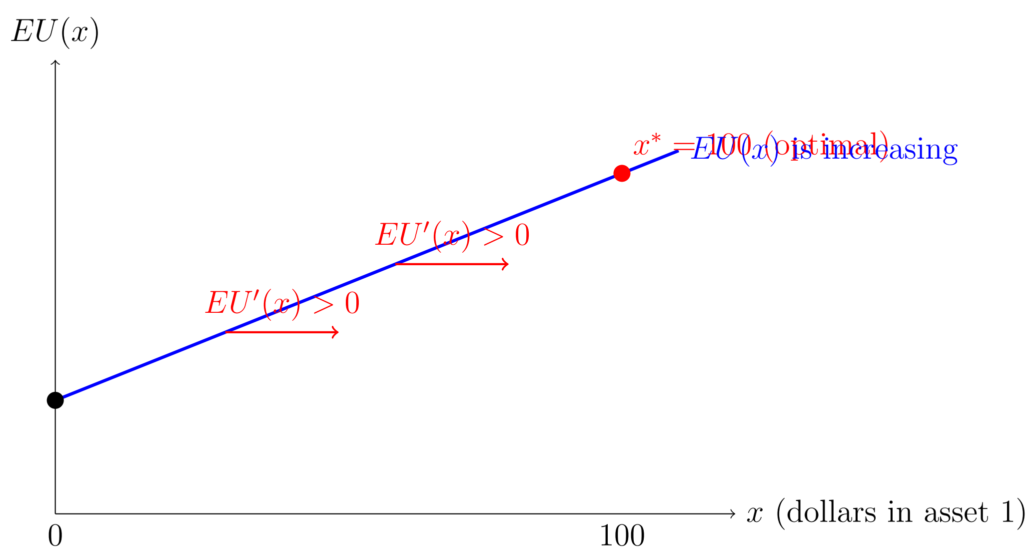
So:

- Increasing x (more in asset 1) $\Rightarrow$ EU increases $\Rightarrow$ good
- Decreasing x (more in asset 2) $\Rightarrow$ EU decreases $\Rightarrow$ bad

Since $EU'(x) > 0$ **even at** $x = 100$, you want to keep going: maximize x.

The maximum possible value is $x = 100$.

Visual Explanation



Since $EU(x)$ is always increasing in x, the maximum is at the rightmost point: $x^* = 100$.

Summary of Cases

Condition	**FOC**	**Optimal x^***
Equation (*) satisfied for some $x \in (0, 100)$	$EU'(x^*) = 0$	Interior solution $0 < x^* < 100$
RHS < LHS even at $x = 100$	$EU'(x) > 0$ for all x	Corner solution $x^* = 100$
RHS > LHS even at $x = 0$	$EU'(x) < 0$ for all x	Corner solution $x^* = 0$

Economic Intuition

Why would RHS < LHS at $x = 100$?

This happens when **asset 1 is so much better than asset 2** that even when you've invested everything in asset 1 ($x = 100$), the marginal benefit of investing even more in asset 1 is still positive.

In other words:

- Asset 1 has better returns or better risk characteristics
- The decision-maker would prefer to invest *more than* \$100 in asset 1 if they could
- But they only have \$100 total, so the best they can do is $x = 100$ (all in asset 1)

Key Takeaways

1. **Interior solution:** FOC gives equality, $EU'(x^*) = 0$
2. **Corner solution:** FOC cannot be satisfied in $(0, 100)$, check boundaries
3. **If RHS < LHS at $x = 100$:** Then $EU'(x) > 0$ everywhere, so $x^* = 100$
4. **If RHS > LHS at $x = 0$:** Then $EU'(x) < 0$ everywhere, so $x^* = 0$
5. **The derivative tells you which direction to go:**
   - $EU'(x) > 0 \Rightarrow$ increase x (want more of asset 1)
   - $EU'(x) < 0 \Rightarrow$ decrease x (want more of asset 2)
   - $EU'(x) = 0 \Rightarrow$ you're at the optimum